02/24

Monday, 24 February 2025 9:30 AM

Midterm → March 7

↳ everything unit lecture 13

↳ Today's lecture.

FD vs. FV

FD
- point/node function
- ↓
- replace derivative/PDE w/ difference function

FV
- Control volume average solution
- ↓
- Has orders of accuracy O^1, O^2
- ↳ Figure/app. fluxes that go into the CV

Infinite no. of FV & FD for a given PDE

Method of lines

→ Space & time are decoupled
→ time difference first
→ & the space
- ↓ → backward / forward difference
- diffusion (central difference)

Convergence → as you refine the grid, solution goes to exact solution

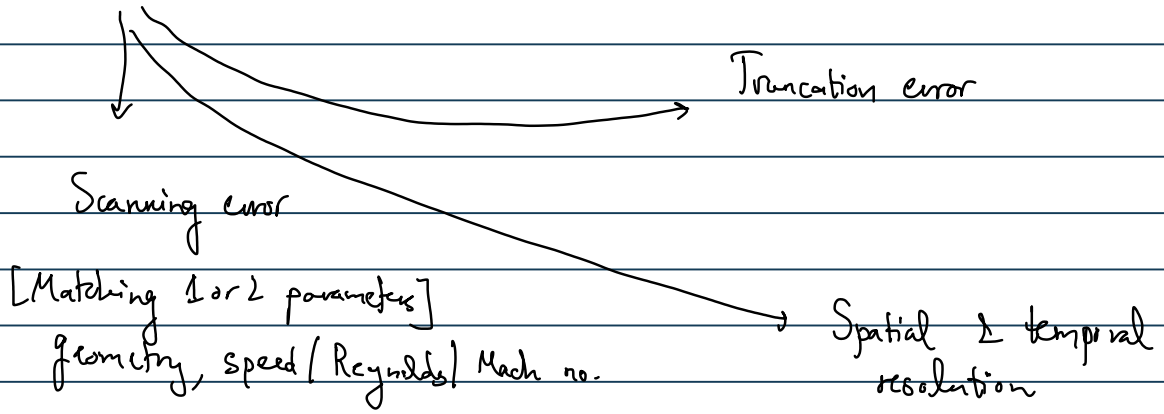
Lax equivalent theorem → linear, well-posed & consistent (unconditionally) & stable then it is convergent

Conservation of errors → guaranteed by FV method

Conservation of errors → guaranteed by FV method
↳ flux coming out from one cell must be the flux in of the next touching cell

Dissipation, diffusion errors.

Errors :



Monotonicity

Positivity

→ To ensure that the oscillations due to discontinuities don't go to crazy